

Akash Reddy Kukkala

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Education

University of Maryland, College Park, US

Master of Engineering, Cybersecurity Engineering

January 2025 – December 2026

GPA: 3.9 / 4

Jawaharlal Nehru Technological University, Telangana, India

Bachelor of Technology, Computer Science and Engineering (Cybersecurity)

September 2020 – May 2024

GPA: 3.5 / 4

Experience

Penetration Testing Intern, Coincent

September 2022 – February 2023

- Performed hands-on penetration tests on 13 intentionally vulnerable web applications and APIs, exploiting OWASP Top 10 vulnerabilities, including SQL injection, XSS and IDOR, using Burp Suite, OWASP ZAP and Metasploit.
- Conducted Nmap reconnaissance and service enumeration to map attack surfaces and identify misconfigured services.
- Authored detailed penetration test reports documenting each vulnerability with reproduction steps, proof-of-concept evidence, business impact and step-by-step remediation guidance, mapped to OWASP Top 10 categories.

Projects

AWS Secure Vault with Threat Detection

In Progress

- Built a serverless encrypted file vault on AWS applying Zero Trust principles with KMS-encrypted S3 storage, enforcing least-privilege IAM roles for every Lambda function and JWT authentication through Amazon Cognito.
- Engineered a threat-detection pipeline combining GuardDuty findings, CloudTrail audit logs and 5 CloudWatch alarms that send real-time SNS alerts on unauthorized access attempts and suspicious API activity.
- Automated the entire stack as Infrastructure as Code with Terraform, making every security control version-controlled, reviewable and reproducible across deployments.

Malware Forensic Investigation

December 2025

- Performed dynamic analysis of three malware variants recovered from a suspect's disk image, capturing outbound HTTP traffic in Wireshark to identify embedded messages and a Base64-encoded encryption key.
- Utilized the decoded key to unlock a VeraCrypt container disguised as an MP3 (flagged by entropy analysis in Autopsy), exposing the final malware payload and concealed files.
- Confirmed malware execution through Prefetch and PylInstaller build artifacts, and documented findings in a formal forensic report with 18 annotated evidence exhibits.

AWS Cloud Security Hardening (E-Commerce Infrastructure)

May 2025

- Assessed an AWS-hosted e-commerce platform across 6 domains, identified 20+ CVSS-rated vulnerabilities including critical wildcard IAM policies(10.0) and a hardcoded database root password(9.1).
- Remediated high-severity findings by enforcing least-privilege IAM roles, restricting public SSH to a trusted IP and encrypting EBS data with KMS, mapping each control to CIS AWS Benchmark and NIST SP 800-53.
- Implemented continuous compliance and visibility with AWS Config rules, account-wide CloudTrail logging to CloudWatch Logs, CloudWatch Agent and automated AWS Backup snapshots.

DDoS Attack Detection System using Machine Learning

April 2024

- Built a machine-learning network intrusion detection system on the NSL-KDD dataset (148K connection records), classifying traffic as normal or DoS, Probe, R2L and U2R attacks with a stacked LSTM model (98.9% accuracy).
- Benchmarked 6 classifiers, with Random Forest reaching 99.5% accuracy, 99.4% recall and a 0.35% false-positive rate; used correlation analysis and RFE to identify SYN error rates and half-open connections as top DoS indicators.

Skills & Certifications

Security Technologies: SIEM (Splunk), EDR (Wazuh), IDS (Snort), Nessus, Wireshark, Nmap, Volatility, Autopsy, The Sleuth Kit, FTK Imager, SIFT Workstation, RegRipper, Plaso, Burp Suite, Metasploit, OWASP ZAP, sqlmap, Ghidra

Analysis & Response: Incident Response (PICERL), Threat Hunting (KQL, SPL), Threat Detection, Log Analysis, Detection Engineering (Sigma, YARA), Disk, Memory & Network Forensics, Chain of Custody, Vulnerability Management

Cloud & Infrastructure Security: AWS (IAM, KMS, VPC, CloudTrail, GuardDuty, Security Hub, Config, CloudWatch, S3, Lambda), CSPM, Least Privilege, IaC (Terraform, CloudFormation), Docker, Kubernetes, VMware, Hyper-V, Linux, Windows

Frameworks: MITRE ATT&CK, NIST CSF, OWASP Top 10, CVSS | **Scripting:** Python (Boto3), Bash, PowerShell, Go, SQL

Certifications: AWS Certified Cloud Practitioner, eLearnSecurity Junior Penetration Tester (eJPT), Microsoft Certified: SC-900, CompTIA Security+ (In Progress)

Research Publications

DDoS Attack Detection using Machine Learning

April 2024

International Journal of Innovative Research in Technology (IJIRT)

Keylogger Detection System

December 2024

International Journal of Innovative Research in Technology (IJIRT)